1. Consider a diagnostic test for a particular disease and the following information:

- 3% of the population has teh disease
- The test correctly identifies a sick patient as positive with probability 95%
- The test correctly identifies a healthy person as a negative with probability 92%

a) Compute the probability that a person has the disease given the test result is positive(theoretical value).

b) Simulate the results of 10^5 tests and estimate, by simulation, the probability requested in the previous paragraph.

c) Compare the results obtained by simulation with the theorical results and comment eventual differences. Comparison and Comments:

2. A comunication system trasnmits bits with an error probability of 0.2. Let X be the random variable representing the number of bits received in error when 16 bits are transmitted.

a) Generate a vector with 10^5 samples of the random variable X. Matlab code:

b) Estimate, from the generated vector, the mean of the X and the variance of X. Mean _____ Variance _____ MAtlab code:

c) Using matlab, compute, theoretically, the probability mass function (PMF) of X and the theoretical values of the mean and variance. Write down the values of some Result: from PMF: P(X=0) = _____ P(X=2) = _____ P(X=4) = _____ P(X=10) = _____ Mean:_____ Variance:_____

d) Compare and Comment the results obtained by simulation with the theoretical results in b) and c).

3. Consider a generator of binary sequences modelled by the discreate time Markov Chain by the incomplete state transition matrix below. Consider the order of the stares represent by the matrix as 0, 1, F, where the state F represents the end of binary sequence.

T = [? 0.5 0 0.25 0.25 ? 0.5 ? 1];

a) Represent in Matlab the complete state transition matrix T inn which the matrix element t_ji corresponds to the transition probability from state i to state j. Represent also the state diagram.

b) Determine the probability (in %) of the sequence 1001 when the generator starts by 1. Result:_____% Matlab code:

c) Determine the probability (in %) of the generated sequence having exactly 5 bits when at the beginning the generator selects the first bit as either 0 or 1 with equal probability. Result: _____% Matlab code: